



GEOMECH INVERSION PLUGIN

User guide v1 – September 2017

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1. INTRODUCTION

For a summary of how to run the inversion tool, see section 5. For known issues, see appendix C.

Geomec models rely partly on assumptions for subsurface input data. For example, the reservoir pore pressure distribution is obtained from a flow model that has many uncertainties, and mechanical properties such as the Poisson's ratio is typically only measured at a few well locations. To calibrate these uncertain geomechanical model predictions, different types of 'hard' data can be used, such as observed deformations from 4-D seismic or measurements of surface subsidence. However, these data can also be used to improve the estimated reservoir properties that go into the Geomec models. Surface subsidence data, for example, can be inverted to estimate reservoir strains or compaction, which can then be used as input for the Geomec model. The inversion is done using fast methods, such as Geertsma for subsidence, that do not require finite element simulations, and are therefore ideal for uncertainty analyses and fast estimation of reservoir properties.

The tradeoff is that most inversion methods use simplified representations of the complex subsurface. Geertsma for example assumes an isotropic linear halfspace for the overburden, which is not realistic when the overburden is faulted or if evaporites are present. Inversion methods do not replace the need for finite element forward models of reservoir and overburden deformation, but by combining the two approaches, the overall modelling process can be improved in terms of simulation times and accuracy. Inversion of hard data provides an objective first-order estimate of reservoir parameters such as pore pressure change or compaction, which is then used as input in a finite element Geomec simulation that captures non-linearities that were not considered in the inversion. The accuracy of the first-order estimate is checked by comparing the residual between the data used for the inversion and the modelled output. As an example, if measured surface displacement obtained from InSAR is used for the inversion, this measured displacement is compared with the displacement modelled in the Geomec model. If a large difference remains between measured and modelled displacement, the residual displacement is inverted in another iteration. The process of iterative inversions combined with forward geomechanical simulations is repeated until the residual is resolved to a minimum (Figure 1).

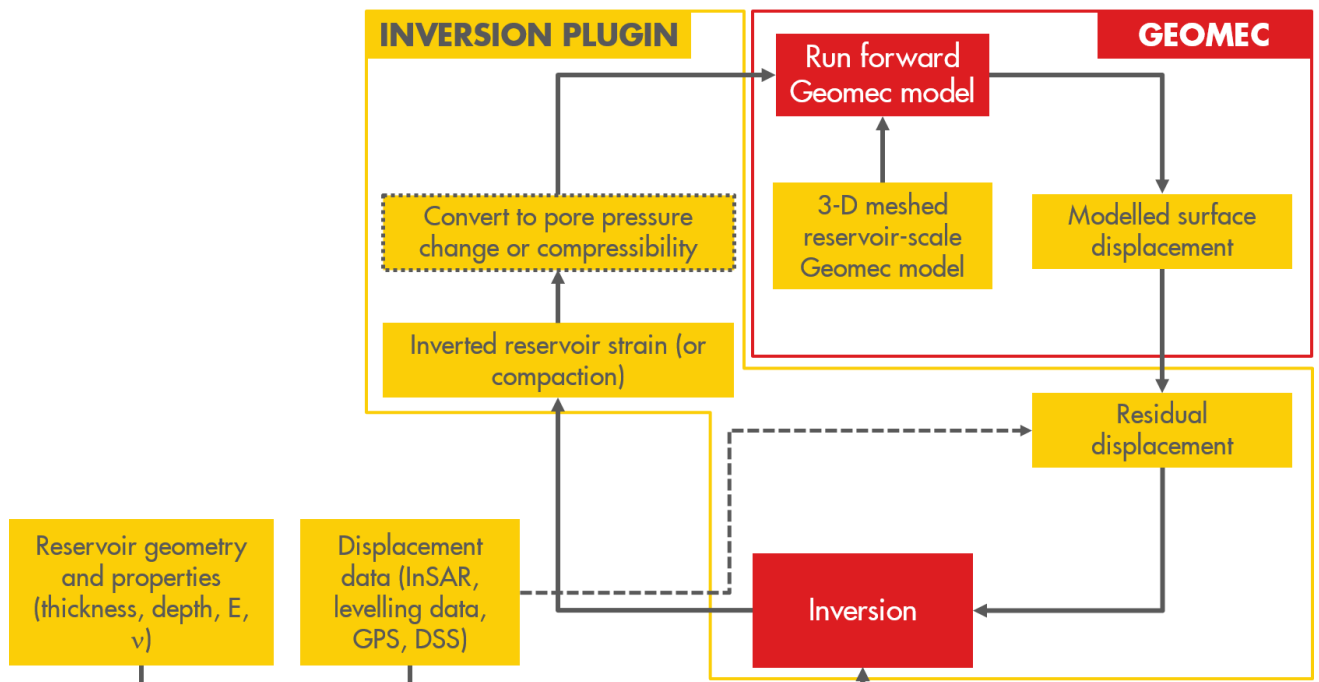


Figure 1 Flowchart for the inversion and forward modelling workflow. Dashed box shows optional steps.

A fast inversion approach has been developed by Hindriks (2017). This inversion scheme is applicable to unstructured grids and able to handle a wide range of data types and sources. It uses the Geertsma solution for an isotropic linear half-space to invert surface displacement to reservoir strain or compaction. The theory behind this inversion scheme is further elaborated in Hindriks (2017). This manual describes only the steps and interactions required to set up and run an inversion process combined with finite element modelling in Geomec. For brevity, we refer to this iterative process of inversion of displacement data followed by a forward finite element simulation as the ‘**inversion-forward model**’ process.

The inversion plugin in Geomec provides a direct link between Geomec and the Inversion Tool (IT, Figure 2), where the inversion can be applied directly to a reservoir model and displacement data obtained from Geomec, and inversion results are returned to Geomec for a forward finite element simulation.

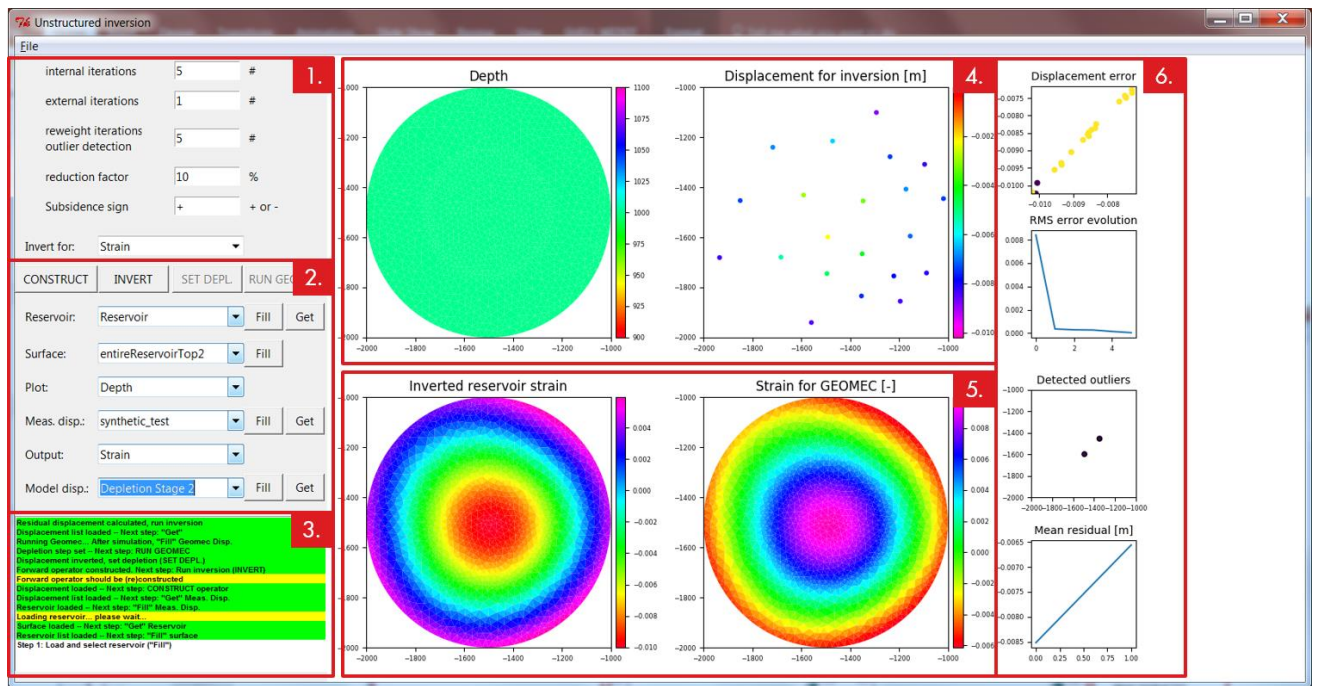


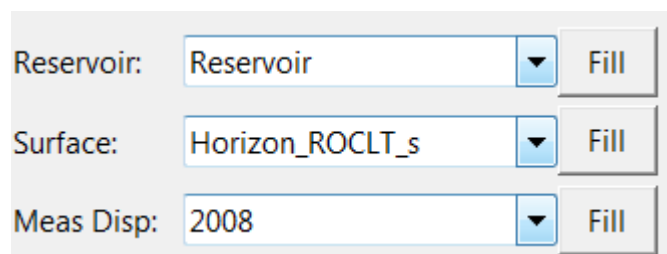
Figure 2 Screenshot of the inversion tool: 1. Settings for inversion; 2. Loading of objects and Geomec interactions; 3. Message dialog providing user guidance and feedback; 4. Display of loaded input data (reservoir and displacement data); 5. Display of inversion results and Geomec depletion steps; 6. Display of QC plots for inversion process.

2. INPUT DATA REQUIREMENTS

The inversion requires the following data to be available:

- Reservoir data (Figure 3): A meshed reservoir model in Geomec with at least one assigned reservoir formation. The thickness of the reservoir is automatically calculated and sent to the IT. Other properties that can be used in the inversion are the Poisson's ratio and Young's modulus, which can be either constant for the entire reservoir or a property distribution assigned to the mesh.
- Top reservoir surface: Although the existence of a reservoir model implies that a top reservoir surface exists, it does require explicit identification of which surface is the top of the reservoir.
- Measured displacement data (Figure 4, Figure 5): 3-D coordinates for each displacement point, and the displacement vector (dx, dy, dz) corresponding to each point. Examples are point clouds with displacement data from LiDAR, levelling data or GPS. Subsiding datapoints should have a positive dz, whereas exhumation has a negative displacement sign, aligned with the conventions in Geomec. The displacement sign conversion can be easily changed by the user in the IT interface if needed (Figure 13). See Appendix B for the format of a displacement data file.

An inversion only requires the reservoir model geometry and the measured displacement data. To execute the iterative process of inversions followed by forward models, a complete Geomec meshed model is required, with one (dummy) depletion step to be able to identify which formation in the model is the reservoir (Figure 6, Figure 7).



Reservoir:	Reservoir	▼	Fill
Surface:	Horizon_ROCLT_s	▼	Fill
Meas Disp:	2008	▼	Fill

Figure 3 Data loading in the IT: Reservoir volume with its properties, top reservoir surface and the relevant time step for measured displacement data.

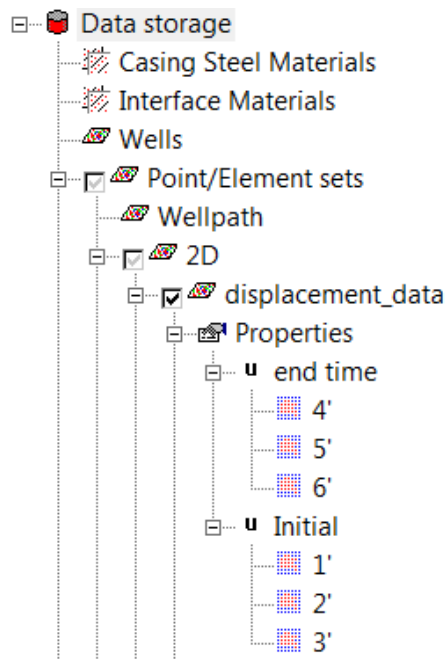


Figure 4 2-D displacement data loaded in the data storage of Geomec. Two time steps are loaded (initial and end time) with each 3 displacement components.

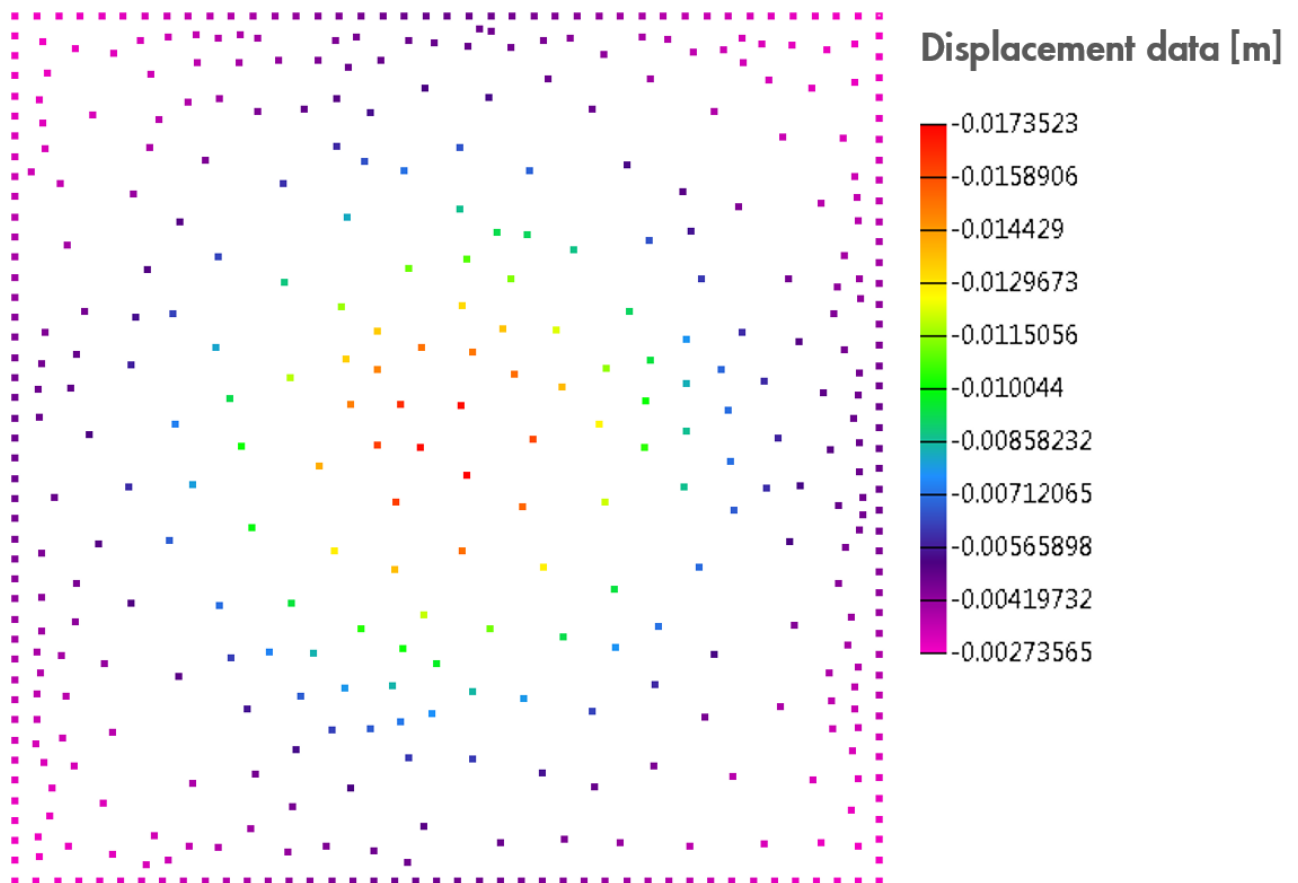


Figure 5 Example of displacement data loaded and visualized in Geomec

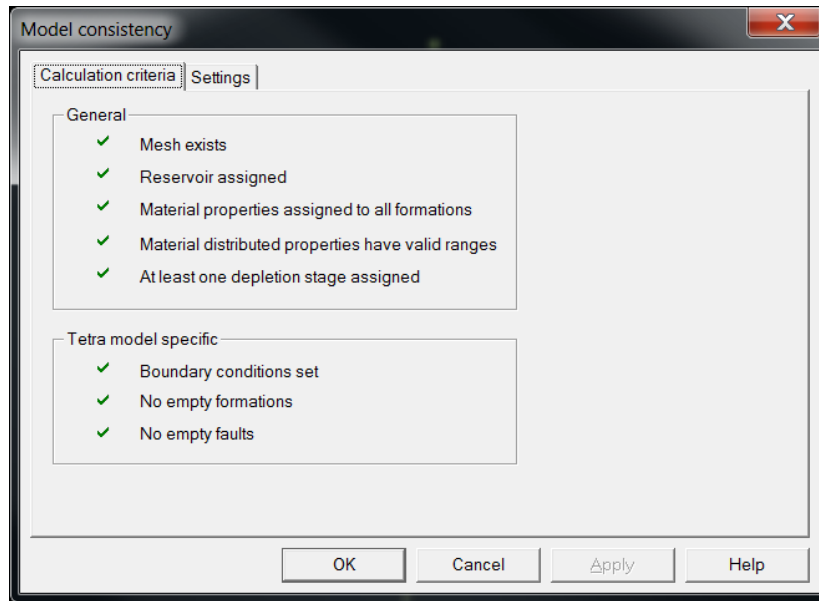


Figure 6 To run the iterative process of inversions and forward simulations, the Geomec model needs to be fully set up, with one (dummy) depletion stage.

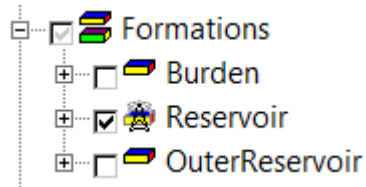


Figure 7 At least one formation should be identified as reservoir.

3. INVERSION

This section describes how to correctly select or load the objects required for the inversion in the IT, the optional settings that can be changed for the inversion and the running of the inversion itself.

3.1. Set-up

As a general note, most loading actions in the IT consist of two steps:

- **Fill:** This action lists/updates the list of all relevant Geomec objects into the IT. After this action, the object of interest can be selected in the dropdown menu.
- **Get:** Once the required object is selected, use this button to load the object and its relevant properties into the IT.

The process of loading objects is split this way to allow for changes to be made in Geomec, e.g. running a simulation, and refreshing the list of available objects in the IT without having to re-open it.

The IT is launched via **Plugins > Non-linear inversion** in the top menu, and initializes in a new window that contains a panel for settings and a panel for plotting data (Figure 8). See Appendix A if the plug-in is not yet installed.

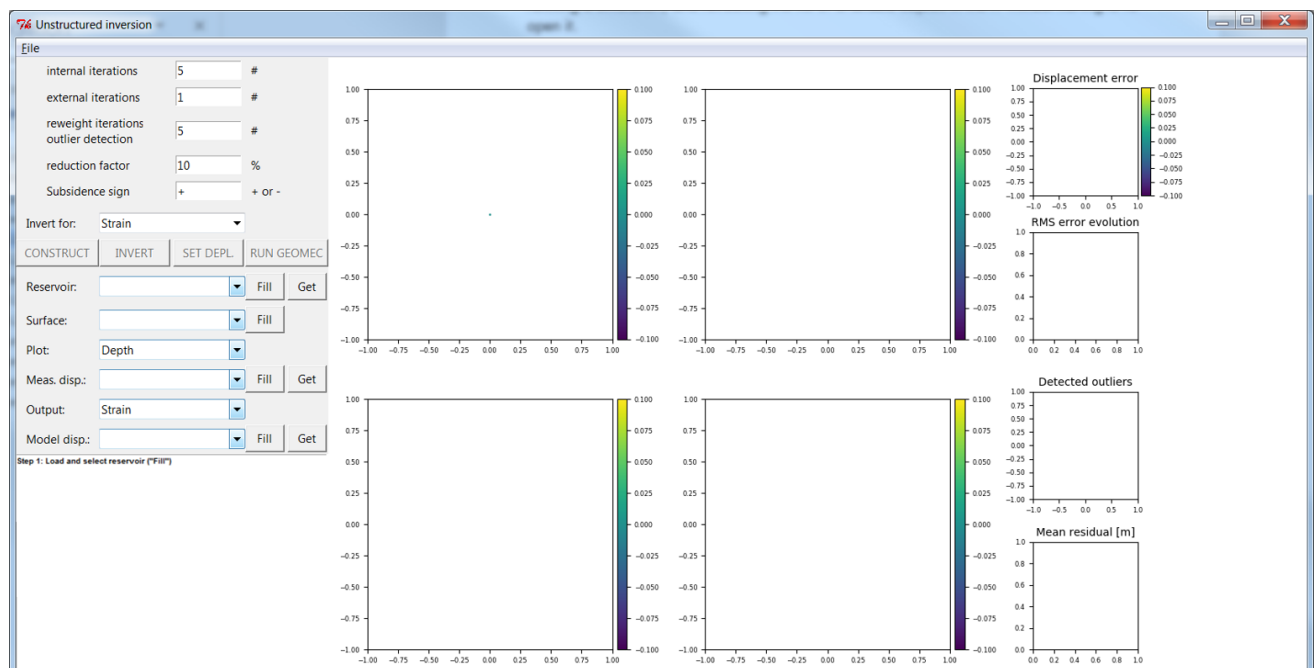


Figure 8 The IT as it looks when launched from Geomec. The pane on the left is for data input and settings. Results are shown in plots on the right.

To load the data required for the inversion, these steps need to be followed:

- Select the reservoir of interest: **Fill – select object** (do not yet click on 'Get') (**step 1** in Figure 9).
- Select the surface that is the top surface of the reservoir of interest: **Fill – select object** (**step 2** in Figure 9)

- iii. Load (**Get**) the reservoir volume and its selected properties (**step 3** in Figure 9). The reservoir geometry and properties are plotted as a function of x, y in the IT (Figure 10).
- iv. Select the relevant measured displacement step: **Fill – select object – get** (**steps 1,2** in Figure 11). Measured displacement as a function of x, y is plotted in the IT.

Figure 9 shows three panels of the Geomec interface for loading a reservoir. Each panel has a 'Reservoir' dropdown, a 'Surface' dropdown, a 'Plot' dropdown, a 'Meas. Disp.' dropdown, and a 'Model disp.' dropdown. Each dropdown has a 'Fill' button and a 'Get' button. The status bar at the bottom of each panel indicates the current step: 'Step 1: Load and select reservoir ("Fill")' for panel 1, 'Reservoir list loaded – Next step: "Fill" surface' for panel 2, and 'Surface loaded – Next step: "Get" Reservoir' for panel 3.

Figure 9 Loading the reservoir in Geomec: 1) "Fill" list with reservoirs; 2) "Fill" list with model horizons; 3) "Get" reservoir geometry and properties (depth, thickness, Poisson's ratio, Young's modulus and initial pore pressure)

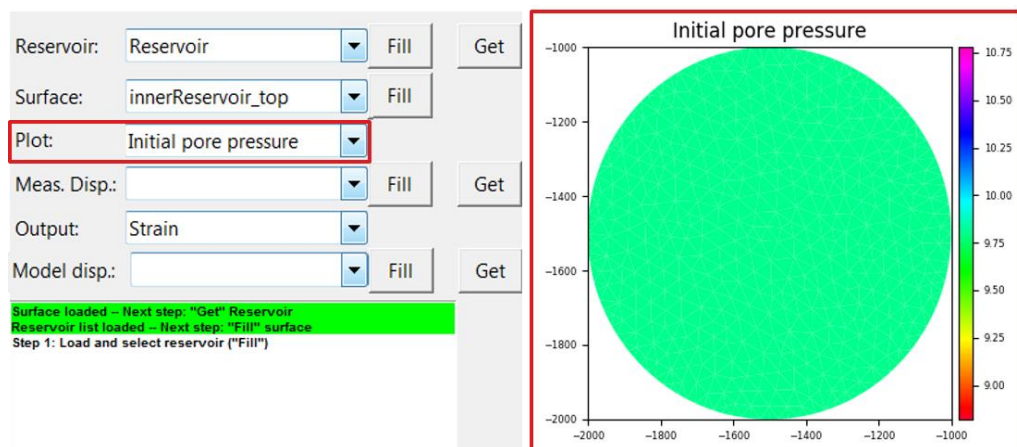


Figure 10 Visualize the top reservoir surface and its properties (depth, thickness, Poisson's ratio, Young's modulus, initial pore pressure)

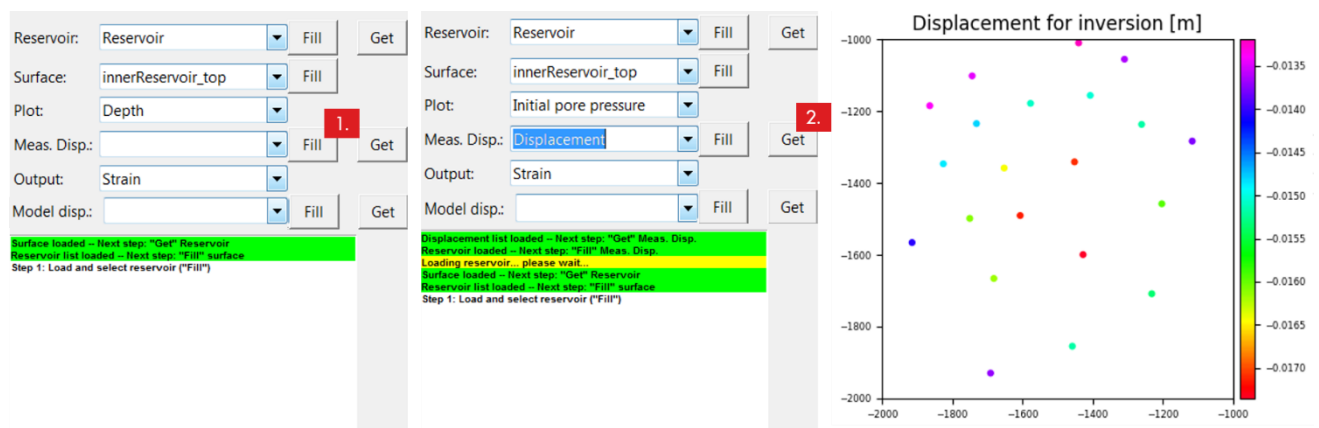


Figure 11 Loading displacement data: 1) Fill the list of measured displacement datasets; 2) "Get" (i.e. load) the selected displacement data

3.2. Execution

The inversion consists of two steps:

- i) Construct a forward operator scheme: Using the reservoir geometry and the distance between the locations of the measurements and the reservoir, a forward operator is constructed for the inversion. See Hindriks (2017) for details. The forward operator only needs to be constructed once, provided that the geometry of the system does not change.
- ii) Execute the inversion: For the given displacements (dx , dy , dz) and the forward operator, displacement data is inverted for reservoir strain (or compaction).

Actions to run the inversion:

1. If reservoir geometry and displacement data are loaded successfully, the IT indicates that the forward operator can be constructed (Figure 12, **step 1** in Figure 13). After successful generation of the forward operator, a confirmation message is given (Figure 14). Note that nothing is plotted at this point. If the top reservoir surface consists of a large number of elements ($>1e5$) this step may take 1-2 minutes.

Prior to executing the inversion, several optional settings can be changed (Figure 13):

- i. Internal iterations: The number of steps to be done in the inversion. More iterations can decrease the inversion error, but too many iterations slows the model down and might result in an overforced/incorrect inversion result.
 - ii. External iterations: An optional additional iteration to ensure that unphysical fluctuations around zero are avoided.
 - iii. Reweight iterations [for] outlier detection: The weight given to iterations.
 - iv. Reduction factor: The amount of data to be removed during outlier detection.
 - v. Subsidence sign: In line with Geomec, subsidence is positive ('+'), whereas exhumation is negative ('-'). The default setting is positive.
 - vi. Invert for [strain, compaction]: By default, displacement data is inverted for reservoir strain. If reliable data on the Poisson's ratio is available, an inversion to compaction can be done. Note that compaction is not directly useable in Geomec simulations, but inverted compaction maps can be exported and used in other workflows.
2. Execute the inversion via the INVERT button (**step 2** in Figure 13). After executing the inversion, the reservoir deformation (strain or compaction) is plotted (Figure 15) and a confirmation message is returned (Figure 14). Three plots for error/uncertainty analysis are generated as well to help judge the accuracy of the inversion (Figure 16).

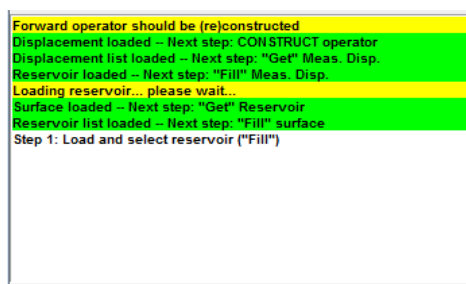


Figure 12 After loading of reservoir and displacement data, the forward operator can be constructed.

internal iterations	5	#	i.
external iterations	1	#	ii.
reweight iterations	5	#	iii.
outlier detection			
reduction factor	10	%	iv.
Subsidence sign	+	+ or	v.
Invert for:	Strain		vi.
1. 2.			
CONSTRUCT		INVERT	
SET DEPL.		RUN GEOMECH	

Figure 13 Inversion settings and buttons for constructing the forward operator (CONSTRUCT) and executing the inversion (INVERT).

```

Forward operator constructed
Forward operator should be (re)constructed
Displacement loaded -- Next step: CONSTRUCT operator
Displacement list loaded -- Next step: "Get" Meas. Disp.
Reservoir loaded -- Next step: "Fill" Meas. Disp.
Loading reservoir... please wait...
Surface loaded -- Next step: "Get" Reservoir
Reservoir list loaded -- Next step: "Fill" surface
Step 1: Load and select reservoir ("Fill")

```

Figure 14 Confirmation message after generating the forward operator.

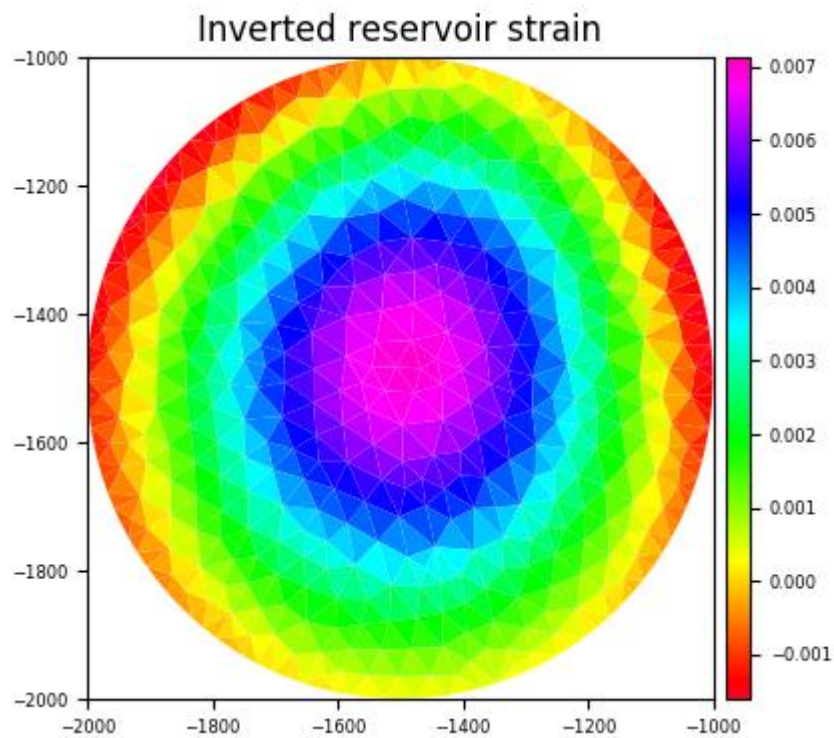


Figure 15 Volumetric strain per element calculated from the inversion



Figure 16 Error plots generated by the inversion: displacement error (measured vs. inverted); RMS error as a function of the number of iterations; x,y coordinates of the detected outlier points.

4. FORWARD MODEL

This section describes how the results from the inversion can be assigned to the Geomec forward model, how the forward model is executed and how results are loaded back into the IT.

4.1. Set-up

After a simulation, the cumulative deformation (as strain, compressibility or change in pore pressure) that is to be used in Geomec as input for the next forward model is plotted (Figure 17). Note that it represents the initial state (e.g. initial pore pressure) plus the change derived from the inversion(s). To select the output type for Geomec and run the forward model, the following steps are required:

- i. Define how the inverted reservoir strain should be sent to Geomec (strain, pore pressure or compressibility) – **step 1** in Figure 18. Note that compressibility is not yet fully supported in Geomec.
- ii. Assign the result from step 1 to a new Geomec depletion step (**step 2** in Figure 18). The new depletion steps can be investigated in Geomec (Figure 19).

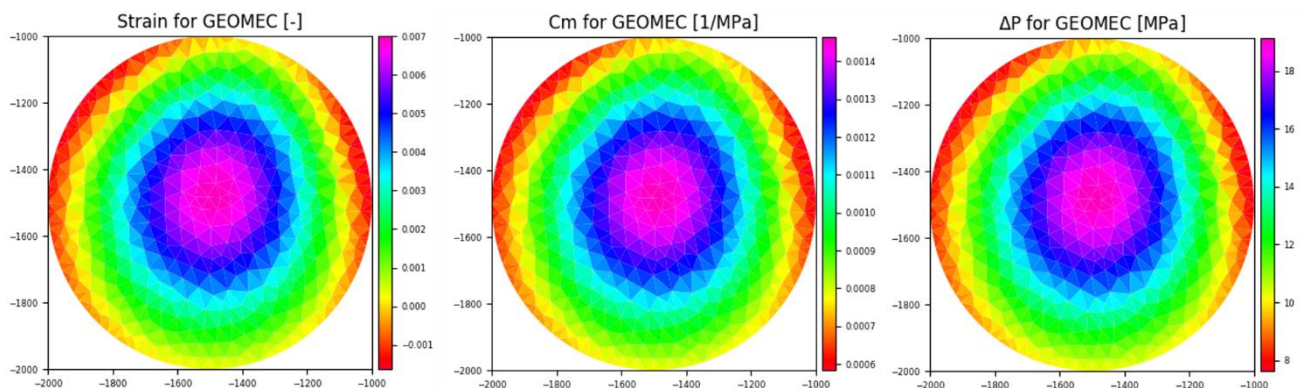


Figure 17 Output from the inversion as a new depletion step for Geomec can be defined as strain, compressibility (Cm) and pore pressure change.

The screenshot shows the 'Invert for' dropdown set to 'Strain'. Below it are four tabs: 'CONSTRUCT', 'EXECUTE', 'SET DEPL.', and 'RUN GEOMEK'. The 'EXECUTE' tab is active. The settings are as follows:

- Reservoir:** 'Reservoir' (dropdown), with 'Fill' and 'Get' buttons.
- Surface:** 'innerReservoir_top' (dropdown), with a 'Fill' button.
- Plot:** 'Displacement' (dropdown).
- Meas. disp.:** 'Pore pressure' (dropdown, labeled 1.), with 'Fill' and 'Get' buttons.
- Output:** 'Strain' (dropdown).
- Model disp.:** (empty dropdown), with 'Fill' (labeled 4.) and 'Get' (labeled 5.) buttons.

Red boxes with numbers 1 through 5 highlight specific elements: 1. 'Pore pressure' dropdown, 2. 'EXECUTE' tab, 3. 'RUN GEOMEK' tab, 4. 'Fill' button for 'Model disp.', and 5. 'Get' button for 'Model disp.'.

Figure 18 Steps for processing the inversion result and assigning these results to the Geomec model.

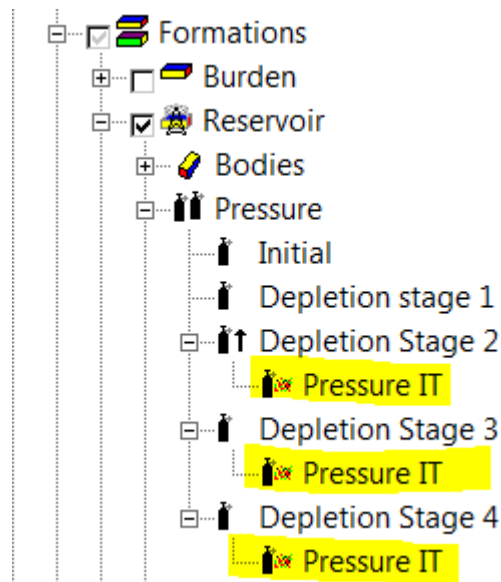


Figure 19 Three depletion states with pore pressures obtained from the inversion tool, labelled as 'Pressure IT'.

4.2. Execution & results analysis

Once the depletion step is successfully set, the Geomec forward simulation can be executed:

- i. Run the updated Geomec forward model (step 3 in Figure 18). Note that running simulations can take some time, and consists of several steps:
 - a. Initializing the model
 - b. Solving for the depletion states (Figure 20)
 - c. Loading results
 - d. Copying the files file (Figure 21)
 - e. Saving the model
- ii. Once the Geomec simulation is fully completed (i.e. there are no more progress bars shown in Geomec), load the updated list of modelled displacement steps (step 4 in Figure 18).
- iii. Load the last modelled displacement result (step 5 in Figure 18). This action calculates the residual between the displacement modelled in Geomec and the original measured displacement, and plots the residual in the IT (Figure 22).

For further QC, the mean of the residual between modelled and measured displacement is plotted as a function of the number of iterations and forward model loops that have been executed (Figure 22c). A message is shown indicating that the IT is ready to execute another inversion (Figure 23).

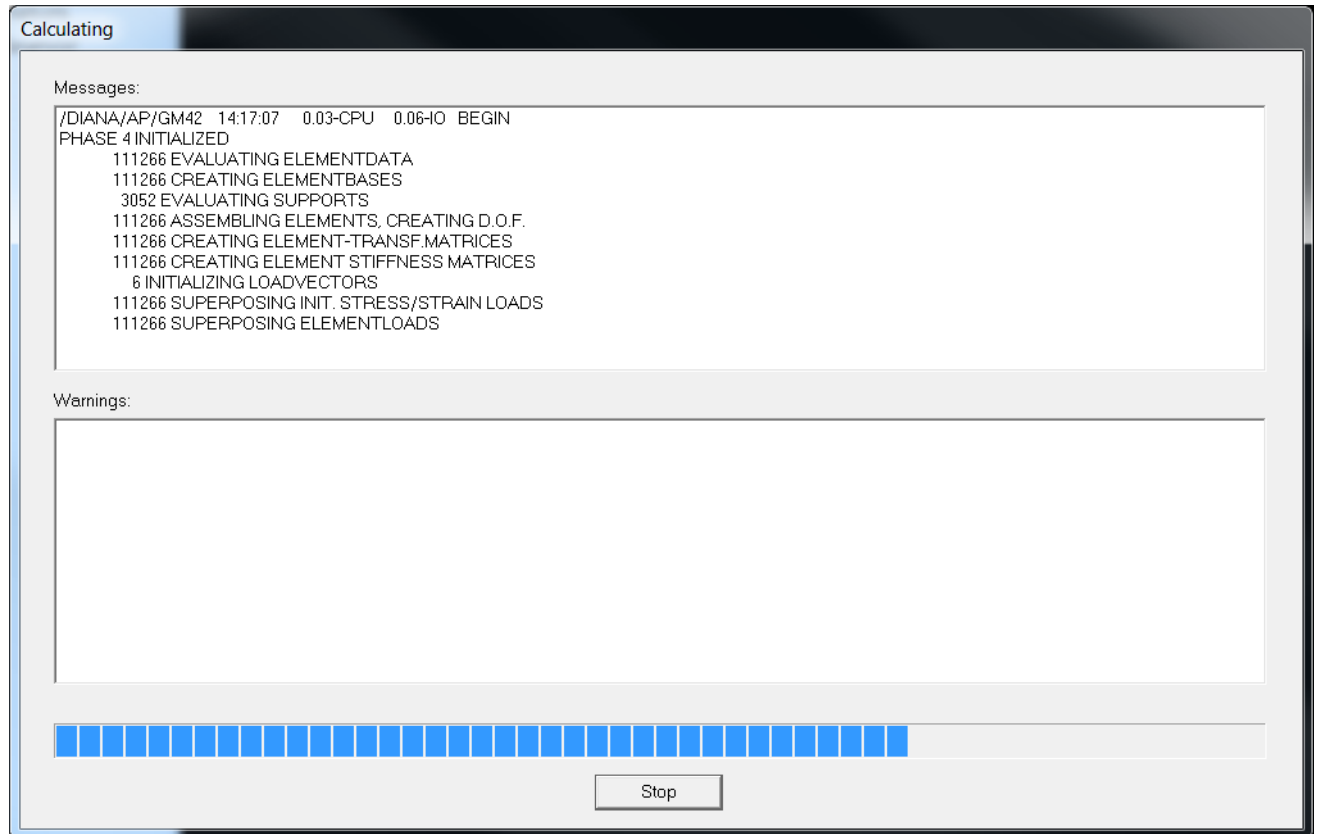


Figure 20 Executing a depletion step in Geomec

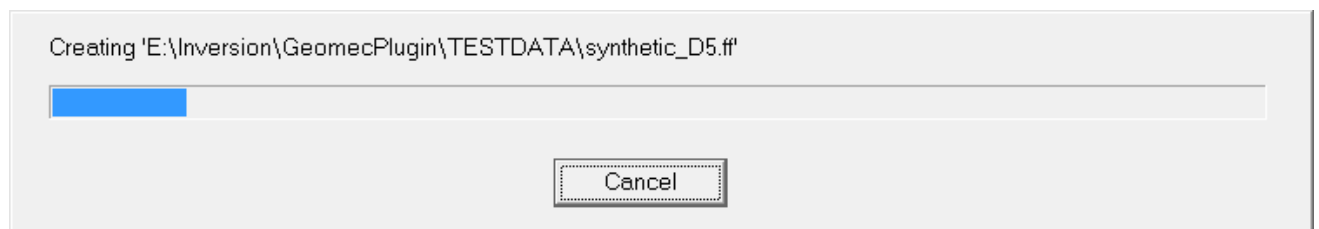


Figure 21 Creating a filos results file

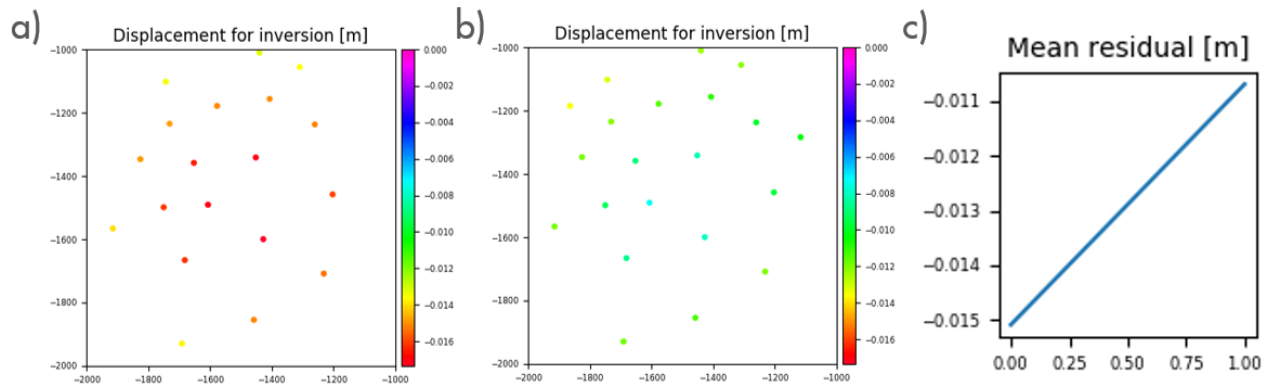


Figure 22 After loading the modelled displacement from Geomec, the residual between the model and the measured displacement data is calculated. The updated residual is shown in the IT (a, b) and the mean of the residual is plotted as a function of the number of iterations and forward model loops that have been executed (c).

```

Residual displacement calculated, run inversion
Displacement list loaded – Next step: "Get"
Depletion step set – Next step: RUN GEOMECC
Inversion executed, set depletion
Forward operator constructed
Forward operator should be (re)constructed
Displacement loaded – Next step: CONSTRUCT operator
Forward operator should be (re)constructed
Displacement loaded – Next step: CONSTRUCT operator
Displacement list loaded – Next step: "Get" Meas. Disp.
Reservoir loaded – Next step: "Fill" Meas. Disp.
Loading reservoir... please wait...
Surface loaded – Next step: "Get" Reservoir
Reservoir list loaded – Next step: "Fill" surface
Step 1: Load and select reservoir ("Fill")

```

Figure 23 If the results from Geomec are successfully loaded, the displacement data used as input for inversion is updated and a new inversion can be executed.

5. SUMMARY & ITERATIVE PROCESS

This section describes in short how several iterations of inversions followed by Geomec forward models are executed, summarizing also the process of loading and running inversions:

1. Open a project with an existing Geomec model.
2. Load an external displacement data file via Import > Import point/elements sets (See section 2).
3. Launch the IT via the top menu in Geomec (Analysis > Inversion Tool).
4. Load the reservoir geometry and its properties (steps 1-3 in Figure 9).
5. Load the displacement data (step 1-2 in Figure 10).
6. Construct the forward operator of the inversion scheme (step 1 in Figure 13).
7. Execute the inversion (step 2 in Figure 13).
8. Assign the inversion result to a new depletion step in Geomec (steps 1-2 in Figure 18).
9. Run the Geomec model (step 3 in Figure 18).
10. Load the modelled displacements – this also updates the displacement in Geomec (steps 4-5 in Figure 18).
11. Repeat steps 7 – 10 to further invert the residual displacement until the residual between modelled and measured displacement is resolved to a level acceptable to the user.

References

Hindriks, K., Alfaro, M.C., Mossop, A., Bisdorn, K. (2017), Robust non-linear inversion for reservoir deformation from surface displacement data. Shell Report.

APPENDIX A: INSTALLING THE PLUGIN

Plugins can be loaded or removed from Geomec via **Plugins > Add** (Figure 24). Plugins are loaded as *.zip files, that contain the plugin source code and a "app.xml file with plugin meta data:

- Name: Name of the plugin to appear in the plugin menu;
- Version: plugin version;
- Manual: path to the manual document;
- Module: name of the python file (without extension) to be executed;
- Function: the function within the module to be executed.

After opening the *.zip file, the plugin should be installed automatically into the plugin installation folder: %APPDATA%/Roaming/Geomec/plugins/**plugin**.

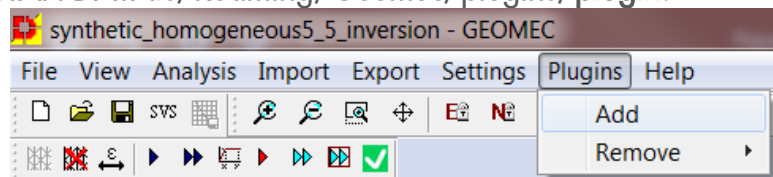


Figure 24 Add, remove and launch plugins via the Plugins menu.

```
<?xml version="1.0"?>
<plugin>
  <name>Non-linear inversion</name>
  <version>1.0.0</version>
  <manual>inversion_plugin_v1.pdf</manual>
  <python>
    <module>inversion_GUI</module>
    <function>main</function>
  </python>
</plugin>
```

APPENDIX B: DISPLACEMENT DATA FILE STRUCTURE

Displacement data is loaded as 2-D or 3-D point data, in a format similar to e.g. pore pressure data. Below is an example file. A few details that need to be considered:

- Data specified under dx, dy, dz represent the change or displacement that occurred between that time and t-1. No absolute values should be used.
- Negative displacements (i.e. subsidence) are **positive** numbers, exhumation/upheave is **negative**, following the same convention as Geomec uses.
- If displacement at a location is specified as 0, it is assumed that this point stays fixed. If data is not available, it should be labelled as NaN rather than 0.
- All units are in [meters].

POINTSET

UNIT SI

DEPL.STAGES

D0	12	2016	Time steps
D1	12	2017	

PROPERTY	Displacement	1'	2'	3'	time0	Names of each time step	
PROPERTY	Displacement	4'	5'	6'	time1		
		dx, dy, dz at t = 0			dx, dy, dz at t = 1		
Northing	Easting	1'	2'	3'	4'	5'	6'
2891811.031	323596.5586	0	0	0	NaN	NaN	0.023268
2892792.487	322596.3213	0	0	0	NaN	NaN	0.04419
2893616.362	321715.9425	0	0	0	NaN	NaN	-0.053695
Y, X coordinates for each datapoint (Z assumed 0)	8845	0	0	0	NaN	NaN	-0.076538
	8213	0	0	0	NaN	NaN	-0.034879
2894203.406	321178.3917	0	0	0	NaN	NaN	-0.077902
2893912.018	320670.8202	0	0	0	NaN	NaN	-0.044614
2893912.018	321504.6405	0	0	0	NaN	NaN	-0.089669
2894195.786	321751.0314	0	0	0	NaN	NaN	-0.070985
2894724.005	321452.748	0	0	0	NaN	NaN	-0.058188
2895034.901	320332.4922	0	0	0	NaN	NaN	0.013496

END

APPENDIX C: KNOWN ISSUES/LIMITATIONS

In the current version of the IT, there are some known series of actions that can lead to unstable behaviour of Geomec, and some features that are not yet implemented:

- Several iterations of inversions and forward models may result in generation of several large temporary simulation (filos) files. For this reason, it is recommended to set the TEMP directories in Settings->Options to some local folder not on the C drive.
- In one Geomec session, opening a different project in some cases results in not being able to launch the IT plugin anymore. In those cases, close and restart Geomec to launch the IT again.
- Displacement data can only be loaded ("Get") after a reservoir has been loaded into the IT, since the displacement data is mapped to the reservoir grid.
- Displacement data outside of the reservoir grid is not taken into account in the inversion process.
- Returning the inversion result as 'Compressibility' to Geomec is not yet supported. Supported outputs are 'pore pressure' and 'strain'.
- The use of strain as a depletion step in Geomec should be used with caution: Inverted strain is returned to Geomec as 'volumetric strain', which is currently assigned as a concentration load in Diana. However, it was argued by Peter Fokker that this would lead to inaccurate stress solutions, and that instead strain should be defined as a strain tensor load. Conversion of volumetric strain to a strain load is however not yet implemented. Currently, inverting for volumetric strain works, but the result is slightly different from an inversion using pore pressure (Figure 25).

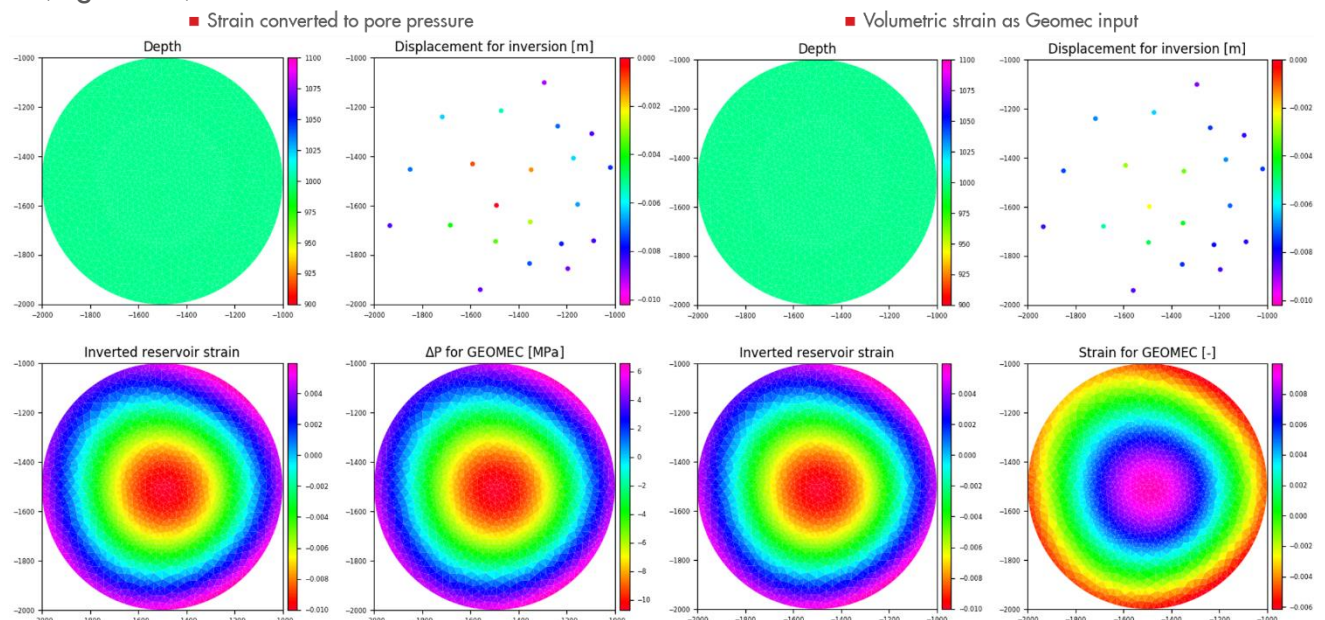


Figure 25 Results of an inversion (inverted strain is identical) sent to Geomec as a pore pressure depletion step (left) and as a volumetric strain depletion step (right) yields small differences in the inversion result, particularly in the center of the subsidence bowl.

- The inversion routine adds the inverted pore pressure to the initial reservoir pore pressure. It is not yet possible to add the inversion result to an existing depletion step.

APPENDIX D: EXAMPLE APPLICATION

The workflow is illustrated using a synthetic model of a depleting reservoir. The model consists of a disk-shaped reservoir with a constant thickness of 10 m and a radius of 500 m. It is divided into two zones. The top of the reservoir is at 1000 m depth. The reservoir is embedded in a 3000x3000x3000 m rock volume (Figure 26). The reservoir and surrounding rock have a strong contrast in mechanical rock properties (Table 1). Both reservoir zones are initially at hydrostatic pressure (10 MPa pore pressure), but the inner part depletes to 2 MPa, whereas the outer part depletes to 5 MPa. The resulting subsidence bowl is modelled in a forward Geomec model. Next, the subsidence bowl is exported as a sparse dataset of surface coordinates and dx, dy and dz displacements, and loaded into the IT. Three loops of inversions and forward models are executed, where compressibility is given, but pore pressure changes are assumed to be unknown. The inverted strain is converted to pore pressure to recreate the depletion profile and subsidence bowl (Figure 27). The absolute values are matched to the original model, though the sharp contrast in pressure zones in the original model is more graduate in the inversion process.

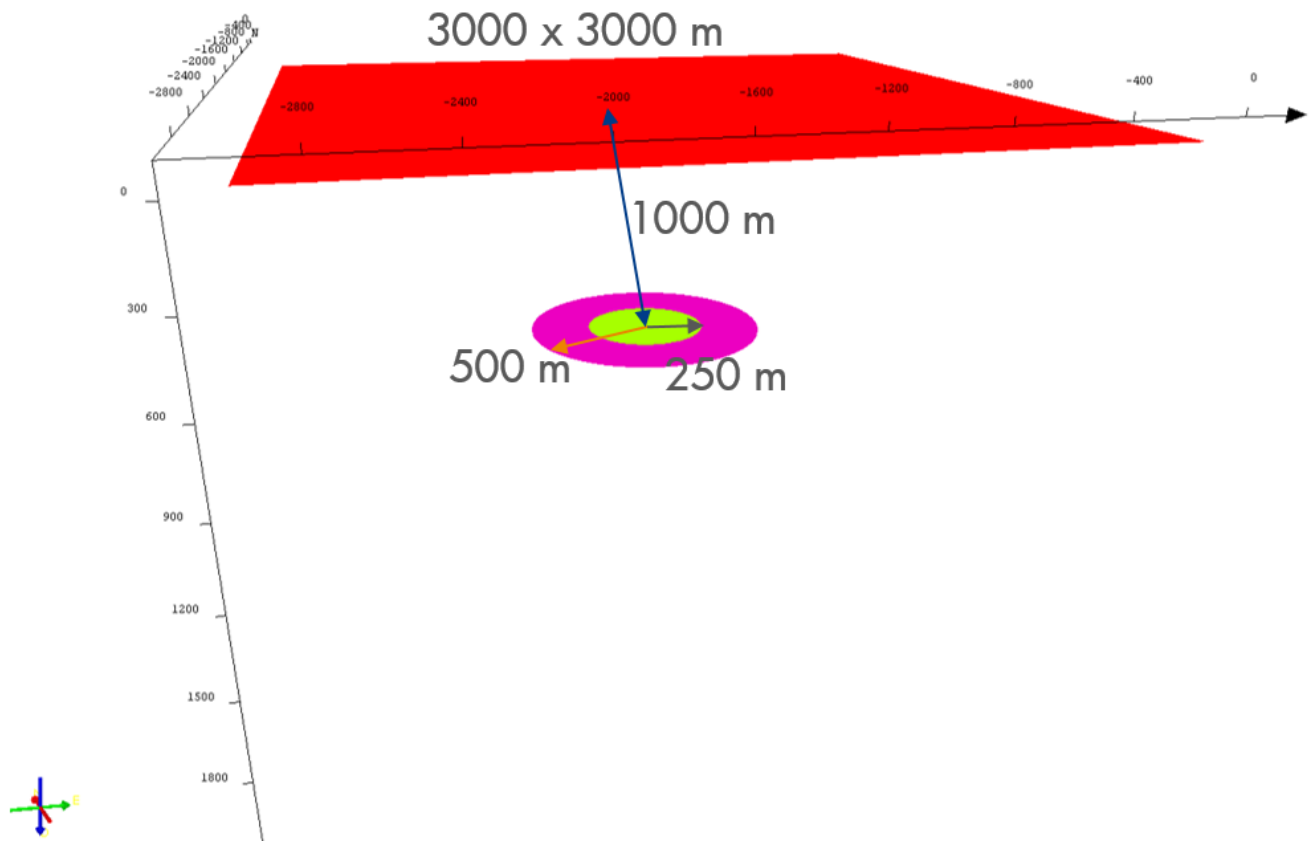


Figure 26 3-D view of the model geometry. Reservoir in pink, top surface of the model in red, base in gray.

Table 1 Summary of relevant properties

Property	Value
Depth (top reservoir)	1000 m
Reservoir width	1000 m
Reservoir thickness	10 m
Young's modulus – Reservoir	2 GPa
Poisson's ratio – Reservoir	0.4

Young's modulus – Overburden	20 GPa
Poisson's ratio – Overburden	0.2
Initial pore pressure	Hydrostatic (10 MPa at top reservoir)

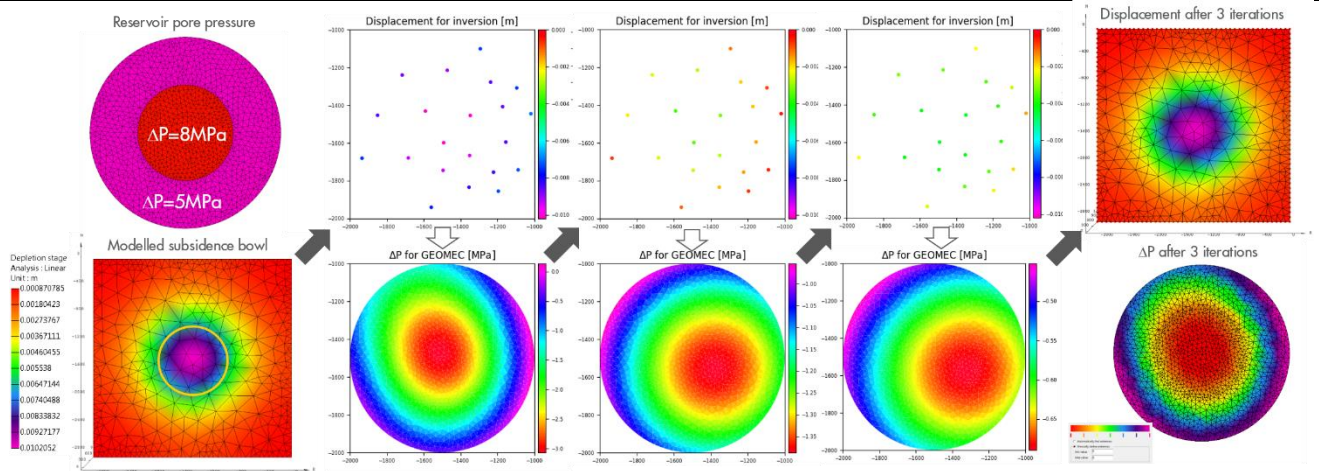


Figure 27 Inversion process and results. Left: initial input obtained from the forward model. Displacement data is inverted to obtain a change in reservoir pressure, which is put in a forward Geomec model. The subsequent residual between measured and modelled data is inverted again.